P-D-1800A
June 1, 1977
SUPERSEDING
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August 7, 1973

FEDERAL SPECIFICATION

DISHWASHING COMPOUND, MACHINE (GRANULAR, BEAD OR FLAKE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers free-flowing, solid form dishwashing compounds suitable for use in spray-type commercial dishwashing machines equipped with or without automatic detergent dispensing equipment.

1.2 Classification. Dishwashing compound shall be of the following types and classes, as specified (see 6.2).

Type I - Non-chlorinated
  Class 1 - With phosphate.
  Class 2 - Without phosphate.

Type II - Chlorinated
  Class 1 - With caustic.
  Class 2 - Without caustic.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

RR-S-366 - Sieve, Test.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-1163 - Box, Corrugated Fiberboard, High Compression Strength, Weather-resistant, Wax-resin Impregnated.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, PA, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Texas, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 1500 - Federal Hazardous Substances Act Regulations.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 460 - Sampling and Chemical Analysis of Soaps and Soap Products.
D 800 - Chemical Analysis of Industrial Metal Cleaning Compositions.
D 1126 - Hardness in Water.
D 2022 - Sampling and Chemical Analysis of Chlorine-Containing Bleaches.
E 70 - pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The dishwashing compounds shall be free-flowing, spray-dried beads, or drum-dried granular, or flake materials formulated from alkaline salts, chelating agents, chlorine releasing agents and other materials, as necessary, to provide products which will meet all requirements of this specification. The dishwashing compounds shall be capable of totally removing food residues, greasy films, and spots from tableware, trays, and glassware when used according to the manufacturer's recommended conditions.

3.1.1 The dishwashing compounds shall be unit packaged in 2-pound (0.907 kilogram) bags (pouches). The dishwashing compounds shall contain no calcium hypochlorite or trichloroisoyanuric acid.

3.2 Composition. The dishwashing compounds shall conform to the requirements shown in table I, when tested as specified in 4.3.2.

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TABLE I. Chemical and physical requirements

	Type I. Percent by weight				Type II. Percent by weight			
	Class 1		Class 2 1/		Class 1		Class 2	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Moisture content	--	5.0	--	5.0	--	5.0	25.0	30.0
Orthophosphate (as P_2O_5)	--	4.5	--	None	--	4.5	--	4.5
Total phosphates (as P_2O_5)	--	20.0	--	None	--	20.0	--	20.0
Other phosphates (as P_2O_5)	16.0	--	--	None	16.0	--	16.0	--
Total silicates (as SiO_2)	4.0	17.0	--	--	4.0	17.0	13.0	15.0
Available Cl_2	--	None	--	None	1.0	2.2	1.5	2.2
Total alkalinity (as Na_2O)	35.0	45.0	--	45.0	--	45.0	--	30.0
Active alkalinity (as Na_2O)	20.0	30.0	20.0	--	20.0	30.0	17.0	--
Free alkali (as NaOH)	15.0	30.0	15.0	30.0	15.0	30.0	--	Nil
Particle size								
Retained on No. 16 sieve	--	10.0	--	10.0	--	10.0	--	10.0
Retained on No. 170 sieve	90.0	--	90.0	--	90.0	--	90.0	--

1/ Acceptable builders shall be used as an alternate to tripolyphosphate in sufficient quantities to pass the water softening capacity test specified in 4.4.13.

3.3 pH and buffer capacity. When the dishwashing compounds are tested as specified in 4.4.12, a solution of each type of compound shall have an initial pH of not less than 11.0 and not more than 12.5. The addition of 10.0 ml of 0.1 N hydrochloric acid shall not reduce the pH of each solution to less than 6.0.

3.4 Water softening capacity. When the dishwashing compounds are tested as specified in 4.4.13, the hard water shall be softened completely.

3.5 Foam height. The foam height of the compounds in distilled water shall be not greater than 1.5 centimeters (5 second value) when tested as specified in 4.3.2.

3.6 Soap. The dishwashing compounds shall show no evidence of soap when tested as specified in 4.4.10.

3.7 Labeling. The containers shall be labeled in accordance with the Federal Hazardous Substances Act. Each container of dishwashing, compound of specified type, shall be durably and legibly marked with the following information, precautions, and directions for use:

FOR MACHINE WASHING OF TABLEWARE, FLATWARE, GLASSWARE, AND TRAYS.

3.7.1 Warning label. All products shall be labeled with the following warning information:

WARNING: INJURIOUS TO SKIN AND EYES
HARMFUL IF INHALED OR SWALLOWED

May contain alkaline substances. Avoid contact with eyes and mucous membranes and prolonged contact with skin.
In case of external contact, flush with water.
For eye contact, flush with water for 15 minutes. Get prompt medical attention.
If swallowed, give large quantities of water or milk. Call a physician.
Keep out of the reach of children.

3.7.2 Caution label. All products shall be labeled with the following caution information:

CAUTION: Alkaline Powder.
Do not mix with acid or other chemicals.
Due to the strength of alkaline components, caustic products are not for use on aluminum ware. Non-caustic products may be used on aluminum ware according to manufacturer's directions.

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3.7.3 Use labels. The manufacturer's "Directions For Use" shall be clearly shown on the label of each container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3 Examination of the end item. The end item shall be examined in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQLs) set forth below.

4.3.1 Examination of the end item container for defects in appearance of dishwashing compound. The sample unit for this examination shall be one filled, primary container. The general inspection level shall be S-4 with an AQL of 2.5 expressed in terms of percent defective.

Examine	Defect
Appearance of dishwashing compound	Not uniform. Not free-flowing, form not as specified. Presence of dirt or foreign matter. Any evidence of segregation of constituents; lumping or caking. Any rancid, fishy or acrid odor.

4.3.1.1 Examination of preparation for delivery requirements. An examination shall be made to determine compliance with the requirements of section 5. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

Examine	Defects
Container	Not as specified.
Contents	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Component missing.
Workmanship	Bulging or distortion of container.

4.3.1.2 Examination of net weight for the two-pound package. An examination of net weight shall be as follows: The sample unit shall be one filled package. The lot size shall be expressed in two-pound packages (0.907 kilogram). The inspection level shall be S-3 and the acceptable quality level shall be 2.5 expressed as percent defective.

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Examine	Defect 1/ 2/
Net weight for the two-pound (32-ounce) package.	Net weight not less than 32 ounces.

1/ Lot shall be rejected if sample data indicates a lot average net weight less than specified net weight.
2/ Weighing device shall have one-ounce increments of measurement.

4.3.1.3 Examination for closure, waterproofing, and banding of containers. When shipping containers are required to be in accordance with PPP-B-636 or PPP-B-1163, examination for defects in closure, water-proofing and banding shall be in accordance with the appendix of that specification.

4.3.2 Testing two-pound (0.907 gram) sample. The end item shall be tested for characteristics as indicated in table II. Two pounds of material shall be drawn from each of five shipping containers and composited. The test sample shall consist of a two-pound composite. The individual determinations shall meet the requirements. The lot should be considered unacceptable if any test fails.

TABLE II. Instructions for testing

Characteristic	Specification reference		Rqmts. appl. to		Number determinations per composite sample unit	Results reported as	
	Require-ment	Test Method	Indiv. unit	Lot average		Pass or fail 1/	Numerically to nearest 2/
Moisture content	Table I.	4.4.4	X	-	2	-	0.1 percent
Orthophosphate (as P_2O_5)	Table I.	4.4.5	-	X	2	-	0.1 percent
Total phosphates (as P_2O_5)	Table I.	4.4.5	-	X	2	-	0.1 percent
Other phosphates (as P_2O_5)	Table I.	4.4.5	-	X	2	-	0.1 percent
Total silicates (as SiO_2)	Table I	4.4.6	-	X	2	-	0.1 percent
Available Cl_2	Table I	4.4.7	-	X	2	-	0.1 percent
Total alkalinity (as Na_2O)	Table I	4.4.8	-	X	2	-	0.1 percent
Active alkalinity (as Na_2O)	Table I	4.4.8	-	X	2	-	0.1 percent
Free alkali (as NaOH)	Table I	4.4.9	-	X	2	-	0.1 percent
Soap	3.7	4.4.10	X	-	2	X	-
Particle size	Table I	4.4.11	X	-	2	X	-
pH and buffer capacity: Initial value	3.4	4.4.12	-	X	2	-	0.1 pH unit
pH value of acidified mixture	3.4	4.4.12	-	X	2	-	0.1 pH unit
Water softening capacity	3.5	4.4.13	X	-	2	X	-
Foaming	3.6	4.4.14	X	-	2	X	-

1/ If failure is indicated, report either description of failure or numerical point of failure, as applicable.
2/ Test report shall include all values on which results are based.

4.4 Test methods.

4.4.1 Preparation of laboratory sample.

4.4.2 Preparation of stock solution of sample. Prepare a solution of the compound by dissolving 50.0 grams in distilled water which has been freshly boiled and then cooled to 20° ± 1°C and reheated to between 55° and 65°C distrilled water. Cool to 20° ± 1°C, make up to exactly 1 liter with distilled water, and mix thoroughly. Filter the solution, if turbid, through a Whatman No. 40 filter paper discarding the first 100 ml of the filtrate and collecting the remainder of the filtrate in a bottle. The stock solution shall be no more than one day old.

4.4.3 Preparation of stock hard water (250 p.p.m). The hard water shall be prepared by dissolving 4.190 g calcium chloride dihydrate and 2.900 g magnesium chloride hexahydrate in freshly boiled distilled water and diluting to 1 liter.

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4.4.4 Moisture content. Determine the moisture content in accordance with ASTM D 460, sections 12 to 15.

4.4.5 Phosphate species. Determine total phosphates (as P_2O_5), orthophosphate (as P_2O_5) and other phosphates (as P_2O_5) contents of a 4.0 percent (by weight) solution of dishwashing compound.

4.4.5.1 Reagents. Reagents "a" through "c" should be prepared and used in testing phosphate species (see 4.4.5).

(a) Concentrated Nitric Acid.
(b) Ammonium Molybdate-Vanadate Solution-Dissolve 40.0 g of ammonium molybdate $(NH_4)_6 Mo_7O_{24} \cdot 4H_2O$) in 400 ml of water. Dissolve 1.0 g of ammonium metavanadate (NH_4VO_3) in a mixture of 300 ml of water and 200 ml of concentrated nitric acid. Add the first solution to the second solution, mix, and dilute to 1 liter with water.
(c) Phosphate Standard-Dissolve 1.9175 g of reagent grade monopotassium dihydrogen phosphate (KH_2PO_4) which has been previously dried at 105° ± 2°C for 1 hour in a liter of distilled water. Ten milliliters of this solution is diluted to 100 ml with distilled water. One milliliter of this solution is equivalent to 0.1 mg of P_2O_5.

4.4.5.2 Calibration. Pipet 5, 10, 15, 20, and 25 ml of phosphate standard into a series of 100-ml volumetric flasks equivalent to 0.5, 1.0, 1.5, 2.0, and 2.5 mg of P_2O_5, respectively. Add to each flask 25 ml of ammonium molybdate-vanadate solution, mix, and dilute each flask to volume with distilled water. Allow 5 minutes for the color development. Measure the color absorbance at 400 to 420 nm with a filter photometer or at 420 nm in a 10-mm cell with a spectrophotometer using water as the reference solution. Plot milligrams of P_2O_5 on the abscissa and absorbance on the ordinate of linear graph paper. A straight line should be obtained passing through the point of origin. A separate calibration curve must be made for each photometer. Each curve must be checked periodically to ensure reproducibility.

4.4.5.3 Orthophosphate. Dilute 10 ml of a 4.0 percent (by weight) solution of dishwashing compound to 100 ml with distilled water. Pipet 5.0 ml of this solution into a 100-ml volumetric flask. Add about 50 ml of distilled water, 25 ml of ammonium molybdate-vanadate solution, mix, and dilute the flask to volume with distilled water. Allow 5 minutes for the color development. Measure the color absorbance in a 10-mm cell at 420 nm with a spectrophotometer using water as the reference solution. Determine the milligrams of P_2O_5 in the 5.0-ml aliquot by referring absorbance value to the reference curve. Calculate the percent of orthophosphate as follows:

Orthophosphate (as P_2O_5), percent = Milligrams of P_2O_5 from reference curve x 5

4.4.5.4 Total phosphates. Pipet 25 ml of a 4.0 percent (by weight) solution of dishwashing compound into a 250-ml beaker. Add 100 ml of distilled water and 20 ml of concentrated nitric acid. Insert a stirring bar and boil gently on a hot plate, equipped with stirrer, for at least 30 minutes. Add distilled water to the beaker to maintain a volume between 50 to 100 ml. Cool to room temperature, remove the stirring bar, and transfer the solution, quantitatively, to a 250-ml volumetric flask. Dilute to the mark with distilled water. Pipet 2.0 ml of this solution into a 100-ml volumetric flask. Add 25 ml of ammonium molybdate-vanadate solution and dilute to the mark with distilled water. Allow 5 minutes for color development. Measure the color absorbance in a 10-mm cell at 420 nm with a spectrophotometer using water as the reference solution. Determine the milligrams of P_2O_5 in the 2.0-ml aliquot by referring absorbance value to the reference curve. Calculate the percent of total phosphates as follows:

Total phosphates (as P_2O_5), percent = Milligrams of P_2O_5 from reference curve x 12.5

4.4.5.5 Other phosphates. Calculate the percent tripolyphosphate as follows:

Other phosphates (as P_2O_5), percent = Total phosphates (as P_2O_5), percent minus Orthophosphate (as P_2O_5), percent.

4.4.6 Total silicates (as SiO_2). Determine total silicates in accordance with ASTM D 800, sections 18 to 21.

4.4.7 Available chlorine (type II). Determine available chlorine in accordance with ASTM D 2022, sections 30 to 33.

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4.4.8 Total and active alkalinity (as Na_2O). Pipet 50.0 ml of the stock solution of sample as specified in 4.4.2 into a 250-ml beaker. Titrate electrometrically against standard 1.0 normal acid to pH 8.1 for the active alkalinity and to pH 3.4 for the total alkalinity.

$$\text{Total or active alkalinity (as } Na_2O\text{), percent} = \frac{A \times 20 \times 3.1}{W}$$

Where:
A = milliliters of acid required for titration.
W = grams of original sample used.

4.4.9 Free alkali (as NaOH). Weigh 5.0 g to the nearest mg of the dishwashing compound into a 400-ml beaker. Add 200 ml of neutral 95 percent ethyl alcohol, cover the beaker, and heat on the steam bath with frequent stirring and maceration of the dishwashing compound until completely disintegrated. Filter through a Gooch crucible with suction into a 400-ml beaker. Wash the residue in the crucible with hot, neutral 95 percent ethyl alcohol until the filtrate is neutral to litmus. Heat the filtrate to incipient boiling, and titrate with standard 1.0 normal hydrochloric acid or sulfuric acid with phenolphthalein. Calculate the percent of free alkali as follows:

$$\text{Free alkali (as NaOH), percent} = \frac{A \times 4.0}{W}$$

Where:
A = milliliters of acid required for titration.
W = grams of sample used.

4.4.10 Soap. Dissolve 2 g of the dishwashing compound in 100 ml of distilled water. Filter off any insoluble matter. Acidify 5 ml of the filtrate with glacial acetic acid in a test tube. The appearance of a fatty acid precipitate or oil drops indicates the presence of soap.

4.4.11 Particle size.

4.4.11.1 Apparatus. Test shall be made using a single-eccentric type mechanical shaker which imparts to the sieves a rotary motion and tapping action of uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Sieves shall conform to RR-S-366.

4.4.11.2 Procedure. Place a 100-g sample on the coarser screen of the nest of appropriate sieves and screen in shaking apparatus. Continue screening for 5-minute intervals until the weight retained on finer screen is reduced by not more than 0.1 g during successive intervals. Weigh material on finer screen and calculate percent of material falling within the appropriate range.

NOTE: In order to evaluate results properly, excessive humidities should be avoided. In the event of failure, retests shall be conducted at a relative humidity not exceeding 50 percent.

4.4.12 pH and buffer capacity.

4.4.12.1 Apparatus. Tests shall be made as specified in ASTM E 70, except that the temperature shall be at 25° ± 1°C. The pH meter shall be calibrated before use by checking against standard buffer solutions of pH 10.0 and pH 7.0.

4.4.12.2 Procedure. Into a 250-ml volumetric flask, transfer 15.0 ml of the stock solution of the sample (see 4.4.2), dilute to mark with distilled water, and mix thoroughly. Determine the pH of the solution as specified in 4.4.12.1. Pipette 50 ml of this solution into a beaker, add 10.0 ml of 0.1N hydrochloric acid to the beaker, stir the solution well, and determine the pH of this acidified mixture. If the acid is stronger than 0.1N, less than 10.0 ml of it therefore being required, sufficient distilled water shall be added to the beaker containing the dishwashing compound solution so that the total volume of acid and water equals 10.0 ml.

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4.4.13 Water softening capacity.

4.4.13.1 Preparation of solutions for test.

4.4.13.1.1 Type I and type II compounds. Transfer exactly 10.0 ml of stock hard water (see 4.4.3) to a 250-ml volumetric flask. This flask, when subsequently filled to mark, will contain 10 grains per gallon hardness. Add approximately 100 ml of distilled water to the flask and then 20.0 ml of the stock solution of compound (see 4.4.2) to the flask. Fill the flasks to mark with distilled water 20° ± 1°C and mix thoroughly.

4.4.13.2 Procedure. Pipette 100 ml of the solution specified in 4.4.13.1.1 into a clean, dry 250-ml beaker. Into another clean, dry 250-ml beaker pipette 100 ml of solution prepared exactly as specified in 4.4.13.1.1 except that the hard water is omitted. Determine the water hardness in accordance with ASTM D 1126, sections 11 to 16. The blue color, without the addition of EDTA, indicates the water has been softened completely.

4.4.14 Foam height.

4.4.14.1 Equipment.

(a) The stirring motor is a standard Hamilton Beach Model No. 936 mixer or equal, with a stirring blade attached. The motor and blade are removed from its original stand and supported on a ring standard by means of a 250-ml beaker clamp.

(b) The test cell is a glass cylinder 7.0 cm inside diameter and 18.5 cm high made by sealing one end to provide a flat bottom.

(c) The baffle used is a cage constructed as shown in figure 1.

(d) A water bath is used to control the temperature ± 0.5°C, in the test cell.

4.4.14.2 Procedure. Two hundred ml of 0.01 percent stock solution of sample are placed in the test cell and allowed to equilibrate in a 66° ± 2°C bath. The stirrer is centered between the four baffles and clamped in position so that the liquid level is midway between the two blades on the stirrer. The stirring motor is run for 3 minutes at a setting to produce a rated 19,000 rpm, to allow the foam to reach maximum height. The stirrer is stopped at 3 minutes and foam height measured in centimeters after 5 seconds. The foam height is determined by measuring the total height in centimeters of the combined foam and liquid (from the bottom of the test cell) and subtracting the initial height of the liquid measured after agitation. The difference is proportional to the volume of entrapped air and is designated as foam height. Distilled or de-ionized water is required for the above test.

5. PREPARATION FOR DELIVERY

5.1 Packaging (commercial). Dishwashing compounds packaged as specified in 3.1.1 do not require additional packaging.

5.2 Packing. Packing shall be level A or Commercial, as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified (see 6.2), thirty (30) two-pound (0.907 kilo) unit packages of dishwashing compound of one type and class only shall be packed in a box conforming to PPP-B-636, grade V2s, style RSC-L. The box shall be closed in accordance with method V and banded in accordance with the appendix of PPP-B-636.

5.2.2 Commercial. The dishwashing compound shall be packed in shipping containers to assure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification.

5.3 Marking.

5.3.1 Civil agencies. The unit and shipping containers and unit loads (when applicable) shall be marked in accordance with 3.8 and Fed. Std. No. 123.

5.3.2 Military agencies. The unit and shipping containers and unit loads (when applicable) shall be marked in accordance with 3.8 and MIL-STD-129.

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5.4 Unitization. (Applicable to full rail car and truck load shipments only). The dishwashing compound shall be unitized for shipment in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

6. NOTES

6.1 Intended use. The dishwashing compounds covered by this specification are intended for use in commercial-type mechanical dishwashing machines in all water conditions- from soft to extremely hard. The chlorinated compound is intended to alleviate the problems of staining in tableware subjected to heavy soil loads.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Marking document required (see 5.3).

MILITARY INTERESTS:

Custodians:

Army - GL (MCA)
Navy - SH
Air Force - 99

Review activities:

Army - MD
Navy - YD

User activities:

Navy - MC

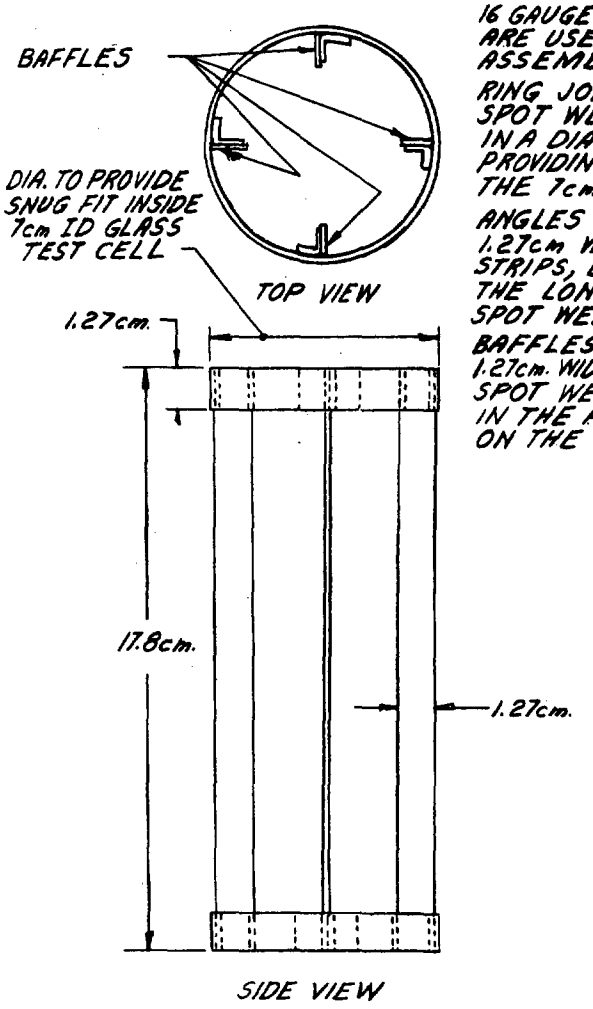
Preparing activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES

VA-DMS
HEW-NIH

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16 GAUGE STAINLESS STEEL STRIPS ARE USED THROUGHOUT THE ASSEMBLY.

RING JOINTS ARE LAPPED, SPOT WELDED AND FEATHERED IN A DIAMETER CAPABLE OF PROVIDING A SNUG FIT INSIDE THE 7cm ID GLASS TUBING.

ANGLES ARE FORMED BY USING 1.27cm WIDE BY 2.58cm LONG STRIPS, BENT IN THE CENTER OF THE LONG DIRECTION AND SPOT WELDED TO THE FORMED RINGS.

BAFFLES ARE FORMED USING 1.27cm. WIDE BY 17.8cm LG STRIPS, SPOT WELDED TO THE ANGLES IN THE POSITIONS INDICATED ON THE DIAGRAM.

FIG 1 BAFFLE CAGE FOR FOAM HEIGHT DETERMINATION

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 35cents each.

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